

orcast: Methodology of Record

orcast

Verified data facts: 2026-06-19

Abstract

orcast is a pilot study on forecast durability for orca encounter probability in the inland waters of the San Juan Islands and the wider Salish Sea (region bounds: latitude 48.40–48.70, longitude -123.25 to -122.75). This document is the methodology of record. It describes a modeled probability surface built from interpretable environmental and temporal kernels estimated as a linear–nonlinear–Poisson (LNP) cascade; the de-biasing logic that reserves continuous acoustic detections for temporal estimation and effort-confounded visual sightings for validation; the live data layer that is wired, ingesting, and validated; the storage and service architecture; the database design; and the leveled estimation/calibration program with explicit fitness gates. The forecast is the default surface throughout and is shown with its own uncertainty; honesty is enforced by what the forecast displays, not by hiding it.

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1 Overview and honest framing

orcast renders a map of where orcas are likely to be encountered, over space and time. The design commitment is that the map heat is a *modeled probability surface*, not a restyled plot of recent detections. The forecast is the default surface and is always shown. Honesty is enforced by what the forecast displays:

1. The forecast always carries its own uncertainty. Early on (few kernels fitted) it is broad and clearly low-confidence; as kernels are estimated it sharpens. The confidence is part of the forecast, not a disqualifying badge.
2. We never render sharper structure than the current evidence supports. The fitness gates (Section 7) govern how confident and how sharp the forecast is allowed to be, not whether it is shown.
3. The detection/validation overlay is always one tap away, so anyone can check the forecast against what was actually observed.

Reported sightings are therefore not the heat: they are the model’s referee (held-out validation), the social/field-log layer, and the citizen-science contribution that feeds estimation.

Status. The data layer of Section 4 is wired, ingesting, and validated (live counts verified 2026-06-19). The kernel model and the studies of Section 7 are a designed, gated program; no kernel is fitted yet. What is live is stated as live; what is planned is stated as planned.

2 The kernel forecast model

2.1 Model form

Model the expected orca encounter intensity at location x and time t as a log-linear (Poisson point-process) product of a spatial habitat term and separable temporal/environmental kernels. In log space the product becomes a sum, so each kernel is additive and separately interpretable:

$$\log \lambda(x, t) = b_0 + s_{\text{space}}(x) + k_{\text{tide}}(\phi_{\text{tide}}(x, t)) + k_{\text{diel}}(h(t)) + k_{\text{lunar}}(m(t)) + k_{\text{season}}(d(t)) + k_{\text{salmon}}(r(t)) + \log E(x, t), \quad (1)$$

where b_0 is the baseline log-rate, s_{space} is the habitat/bathymetry/channel prior, k_{tide} is a cyclic kernel over tidal phase / current state, k_{diel} over solar hour angle, k_{lunar} over moon phase (~ 29.5 d), k_{season} over day-of-year, k_{salmon} an aperiodic smooth over the salmon run-timing index, and $\log E(x, t)$ the observation-effort / detection offset. For display the rate is converted to a 0–1 encounter probability per cell and time bin,

$$p(x, t) = 1 - e^{-\lambda(x, t) \Delta t}. \quad (2)$$

2.2 Why log-linear separable kernels

Each kernel is an interpretable, publishable curve; the form is estimable from sparse data (far fewer parameters than a free 4-D surface); and it is honest about structure, assuming separability that the validation step tests rather than asserts. Equation (1) is, formally, the LNP cascade used to characterize sensory neurons: environment $s(t) \rightarrow$ linear kernels $k \rightarrow$ nonlinearity $f = \exp \rightarrow$ rate $\lambda(t) \rightarrow$ events (detections). A hydrophone is a “detector unit,” its detection times are the “spike train,” and the environmental cycles are the “stimulus.” That correspondence is why the neuroscience estimation toolkit (Section 7) applies directly.

2.3 The kernels and their data

Kernel	Timescale	Data source	Available now?
k_{tide}	$\sim 6.2/12.4$ h	NOAA CO-OPS 9449880; currents PUG1702	Yes (stored)
k_{diel}	24 h	computed from timestamp + lon	Yes (on demand)
k_{lunar}	29.5 d	ephemeris (computed)	Yes (on demand)
k_{season}	annual	computed from date + lat	Yes (on demand)
k_{salmon}	weeks–seasonal	Albion + DART, climatology fallback	Yes (stored)
s_{space}	static	bathymetry + sighting density	Partial

Table 1: Forecast kernels, their covariate timescales, data sources, and current availability.

3 Instruments and the de-biasing logic

Visual sightings are presence-only with severe, structured observation bias: more observers near Lime Kiln, ferry routes, in daylight, and in summer. Naively fitting k_{diel} to sightings would learn “daytime” because people watch in daytime; k_{season} would learn tourism. This is designed out two ways. First, acoustic data carries the temporal truth: OrcaHello / Orcasound hydrophones monitor continuously, with effort that does not track human daylight or tourism, so the temporal kernels are estimated primarily from continuous acoustic detections at fixed stations where effort E is known and roughly constant. Second, an explicit effort offset $\log E(x, t)$ enters the model wherever effort is heterogeneous, so the kernels estimate presence, not observation. This splits the instrument roles: hydrophones estimate the temporal kernels; NOAA tides/currents, ephemeris, and the salmon index are the stimulus covariates; visual sightings (OBIS / community) are held-out validation plus the social and citizen-science layers. Acoustic presence is not the same as a visible encounter; the bridge from acoustic λ to visible-encounter probability is an open decision.

4 The data layer (what is wired)

Time-series history is stored as S3 partitioned NDJSON under `timeseries/{stream}/{station}/{yyyy}/{mm}.nd`. DynamoDB tables serve entity / “latest” reads. As of 2026-06-19 the data gate is satisfied and `/health` reports `healthy` with 76 normalized sightings. Table 2 lists the verified live counts.

Stream / store	Source	Coverage	Records
<code>acoustic_detections</code> (S3)	OrcaHello (<code>haro_strait</code>)	2020-09..2021-04	761
<code>env_water_level</code> (S3)	NOAA Friday Harbor 9449880	hourly history	175,445
<code>env_currents</code> (S3)	NOAA Rosario Strait PUG1702	predictions	17,544
<code>salmon_run_index</code> (S3)	Albion / DART + climatology	daily	783
<code>station_uptime</code> (S3)	9 Orcasound stations	accumulating	27
<code>sightings: obis_verified</code> (DynamoDB)	live OBIS API	live	248
<code>sightings: inaturalist</code> (DynamoDB)	iNaturalist API	live	5
<code>sightings: orcahello</code> (DynamoDB)	OrcaHello	live	1
<code>sightings: community</code> (DynamoDB)	community self-posting	live	1

Table 2: Live data inventory, verified 2026-06-19. All data-wiring fitness gates satisfied.

Static assets (committed): `san_juan_land.geojson` (50 island polygons, OSM coastline) and `san_juan_bathymetry.json` (589 points, NOAA ETOP01, depth -320 to $+621$ m). Diel/solar/lunar

covariates are computed on demand. All adapters write through a single `TimeSeriesStore` abstraction (S3 NDJSON in production, an in-memory mirror for tests): `put_series`, `get_series`, `list_stations`; partitions are merged per month and deduplicated by (t, id) . An `EventBridge` rule (30 min) invokes a Lambda that makes keyed calls to `/api/sightings/ingest` and `/api/timeseries/refresh`.

5 Architecture and data flow

Sources feed adapters, adapters write to the two stores, the API serves status and ingest endpoints, and the kernel model reads the stores to produce the forecast surface (Figure 2). The public community endpoint and the `EventBridge` schedule both write into this pipeline: the public `POST /api/community/sightings` writes a citizen-science submission, and the scheduled service self-writes the time-series streams.

6 Database design and ERD

Two stores back the system: DynamoDB for entities (each table keyed by `pk = id`) and the S3 time-series NDJSON store for the environmental and acoustic streams. The five DynamoDB tables hold `NormalizedSighting` (`orcast-aws-backend-sightings`), `Hotspot` (`-hotspots`), `ProbabilityReport` (`-reports`), `IngestionRun` (`-ingestion-runs`), and `CommunitySubmission` (`-community-submissions`). A `ProbabilityReport` contains `Hotspots`; a `Hotspot`’s `evidence_sighting_ids` reference `NormalizedSightings`; an approved `CommunitySubmission` is adapted into a `NormalizedSighting` (`source = community`); an `IngestionRun` records a `SourceStatus` per source and produces `NormalizedSightings`; and the `acoustic_detections` time-series is derived from `OrcaHello` (Figure 3).

7 Estimation and calibration studies

Recovering a kernel k and a nonlinearity f from a spike train and a stimulus is exactly the LNP problem, so the kernels are estimated with sensory-neuroscience methods. Each level states the data it needs and a go/no-go fitness gate; the forecast is not allowed to sharpen past a level until that level passes on held-out data, with splits blocked by time (whole cycles held out) to prevent leakage.

7.1 Estimation methods

1. **Peristimulus time histogram (PSTH).** For each cyclic covariate, define a repeating stimulus event (sunrise; flood-current onset; new moon; Jan 1 / run onset; run peak), align detections to it, bin by phase, divide by occupancy/effort, smooth with a periodic spline, and bootstrap cycles for confidence bands. The cheapest, most legible first estimate.
2. **Reverse correlation / detection-triggered average.** For continuous covariates (current speed, tide height, temperature), average the covariate trajectory in a window around each detection; whitening the covariate autocorrelation separates correlated covariates such as tide and current.
3. **Joint LNP / Poisson GLM–GAM (the actual estimator).** $\log \lambda(\text{station}, t) = b_0 + a_{\text{station}} + \sum_j \text{spline}_j(\text{cov}_j(t)) + \log E(\text{station}, t)$, with penalized cyclic splines, station random

effects, and an effort offset, fit by penalized likelihood or with GP/spline priors. The fitted splines are the kernels and should match the PSTH/STA shapes.

7.2 Leveled plan with fitness gates

Level	Goal	Fitness gate
L0	Instrument characterization	Per-station effort series known; detector ROC AUC / d' reported with CI
L1	Single-covariate PSTH (diel)	PSTH beats a phase-shuffled null; shape consistent across stations
L2	Joint temporal LNP	Held-out log-likelihood beats climatology and best single covariate; kernels stable across folds
L3	Prey + space $\rightarrow$ field	Held-out skill beats detection-density baseline; calibration within tolerance (PIT $\approx$ uniform)
L4	Population decoding + uncertainty	Localization validated vs held-out visual sightings; beats persistence; calibration holds out of sample
L5	Full-confidence forecast	L3–L4 passed; calibration holds; displayed confidence high

Table 3: Leveled estimation plan. Each gate raises how sharp and confident the forecast is allowed to be, not whether it is shown.

7.3 Sensory-transduction calibration assays

Beyond fitting, the curves earn trust through assays that probe whether the response behaves like a real transduction system: dose-response (threshold and saturation via a Hill-type curve); adaptation/habituation within a presence bout; signal-detection (d' , ROC) for the detector across noise conditions; null and shuffle controls for every PSTH; cross-validation skill (held-out deviance, Brier, log-loss vs baselines); reliability diagrams and PIT histograms; and a population/emergence read-out that treats the network as a distributed sensor array (per-station LNP units feeding a population decoder), framed as sensor fusion rather than a claim of cognition.

8 Roadmap

Phase 0 (now): data layer wired and validated; covariate library; instrument-panel overlay UX; forecast shown prior-driven, broad, low-confidence. **Phase 1:** fit $k_{\text{diel}}, k_{\text{tide}}, k_{\text{lunar}}, k_{\text{season}}$ from ≥ 1 year of acoustic (L1–L2); publish kernel curves with CIs; stand up the validation harness. **Phase 2:** add k_{salmon} and s_{space} ; full $\lambda(x, t)$ (L3); backtest vs baselines. **Phase 3:** richer uncertainty-aware display and population decoding (L4); per-ecotype models if labels support it. A `kernel_model` module computes covariates, holds fitted coefficients, and serves $\lambda(x, t)$ grids; fitting runs offline and the service loads coefficients.

9 Honesty guardrails

orcastr does not run live ML; it consumes the OrcaHello detections API. OrcaHello records are unreviewed acoustic candidates at a named hydrophone (the unreviewed stream is dominated by false positives), and a detection’s coordinates are the hydrophone’s location, not the animal’s. No pod or ecotype identity is shown unless a source field provides it. External research (Perch 2.0, Palmer 2025, Pod.Cast, Watkins) is cited as provenance, not presented as orcastr capability. Licenses are respected (OrcaHello-RAIL for the model; CC BY-NC-SA 4.0 for Orcasound open data; orcastr is non-commercial). The forecast is shown honestly, not hidden: the fitness gates raise its confidence, they do not flip a visibility switch, and the validation overlay is always available so the forecast can be checked against observed detections.

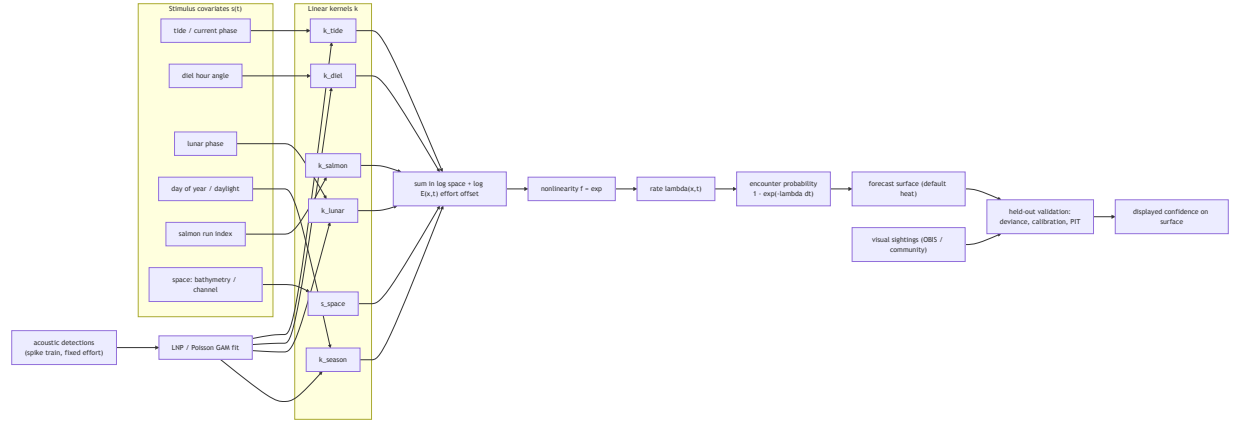


Figure 1: Model pipeline: stimulus covariates drive linear kernels, summed in log space with an effort offset, passed through $f = \exp$ to a rate $\lambda(x, t)$ and an encounter probability. Acoustic detections fit the temporal kernels; visual sightings validate the surface.

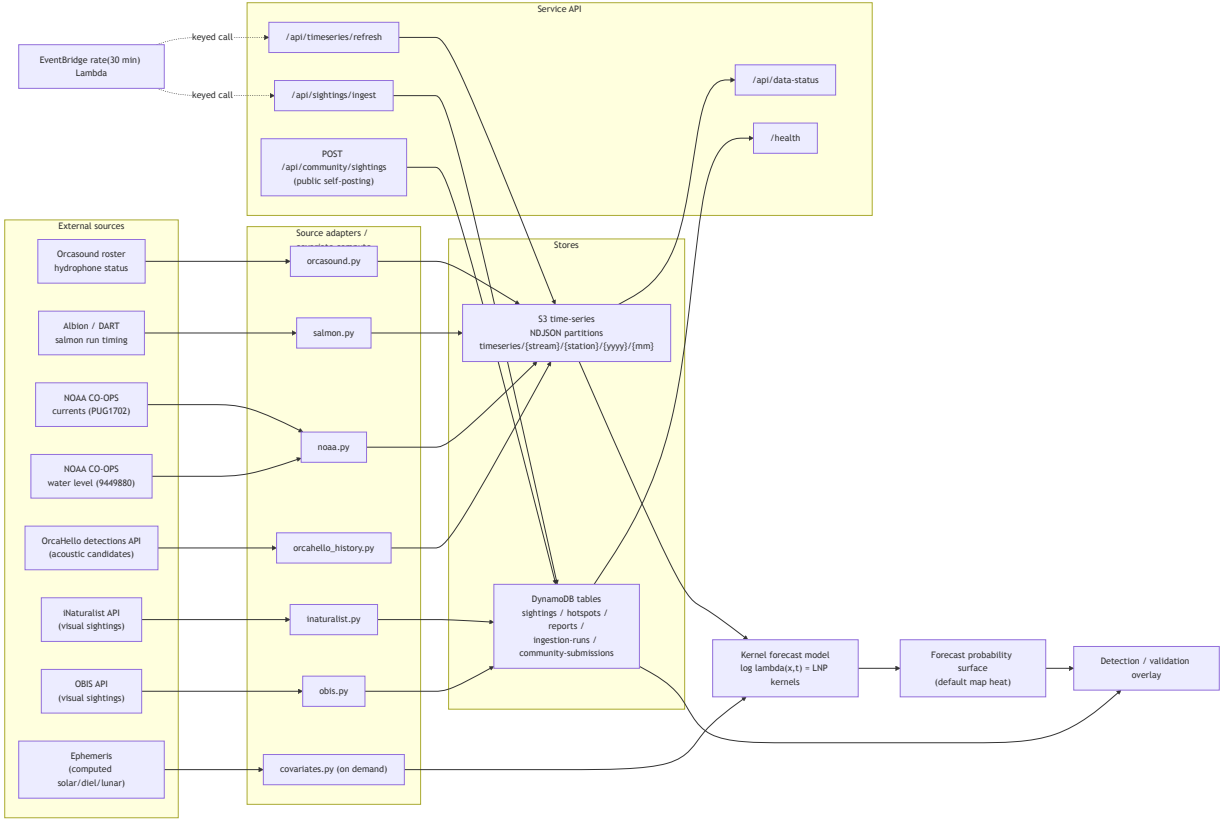


Figure 2: Data-wiring flow: external sources → adapters → S3 time-series and DynamoDB stores → service API → kernel model → forecast surface, with the community self-posting endpoint and the EventBridge schedule writing into the same pipeline.

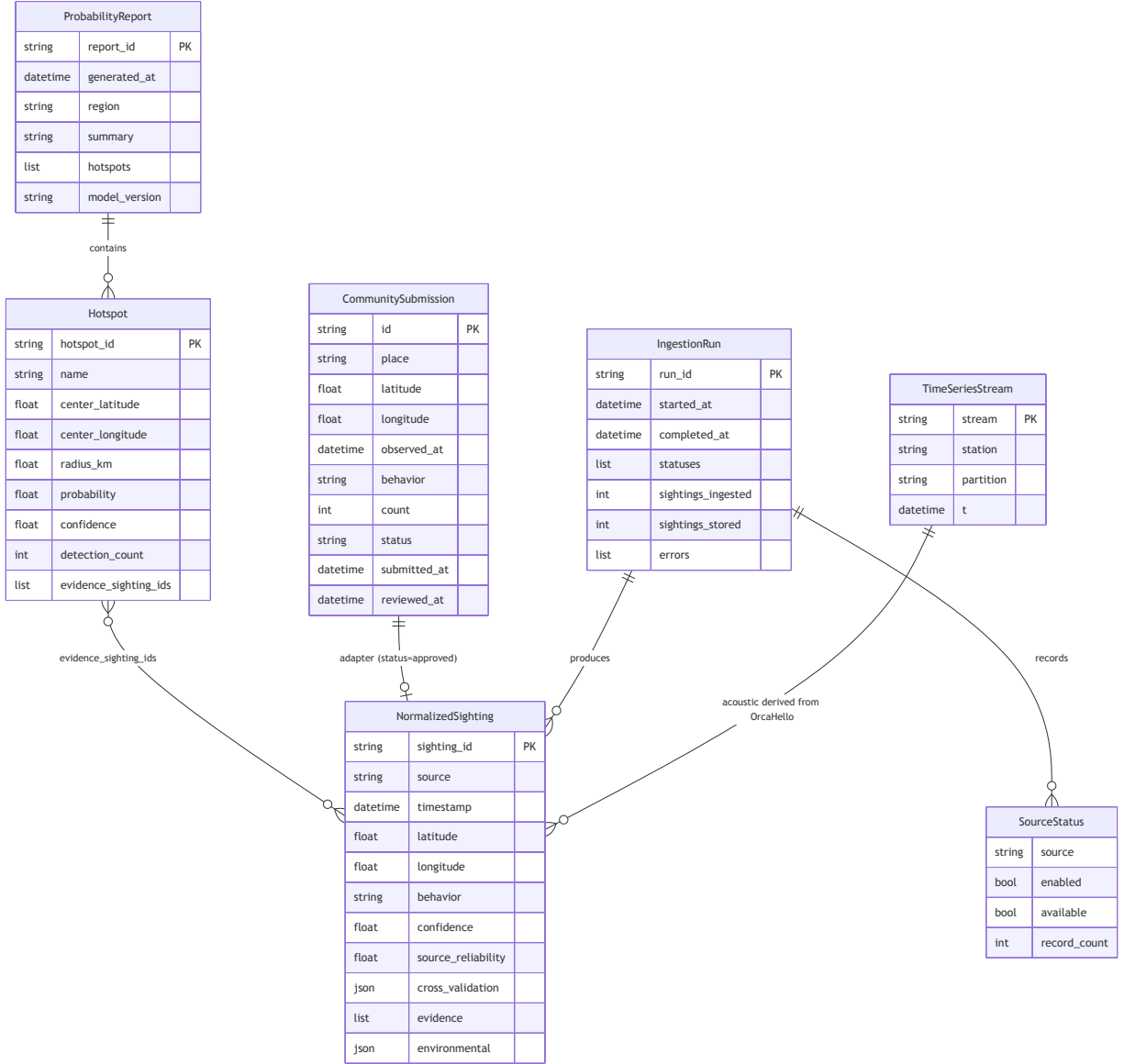


Figure 3: Entity-relationship diagram of the five DynamoDB entities, the **SourceStatus** recorded per ingestion run, and the time-series stream, with the relationships described in the text.